IS1201: Programming & Problem Solving

1. Introduction

Viraj Welgama

IS – 1st Year – Sem I

1	IS 1201	Programming and Problem Solving	2L+1P	
1	IS 1202	Computer Systems	2L	
1	IS 1203	Foundations of Information Systems	2L	
1	IS 1204	Fundamentals of Software Engineering	2L	
1	IS 1205	Introduction to Management	2L	
1	IS 1206	Mathematics for Computing	2L	
1	IS 1207	Internet and Web Technologies	2L+1P	
1	EN 1201	Communication Skills	1P	Yes
1	EN 1202	Application Laboratory	1P	Yes

About…

- Code: IS1201
- Name: Programming & Problem Solving
- Requirement: Core
- No of Credits: 3 (2L 1P)
- Rubric: 70% 30%
- Delivered by:
  - 8/15: Viraj Welgama (wvw@ucsc.cmb.ac.lk)
  - 7/15: Roshan Abeweera (rns@ucsc.cmb.ac.lk)

Practical

- Wednesday 1-3pm
- May be considered as one of the assignment

- Instructors:
  - Ms. M.M.A. Devindi (ADM)
  - Ms. P.M.U. Sandamini (UPE)

Course Objectives

- LO1: Understand the overview and history of programming languages and different programme design and development approaches.
- LO2: Understand and apply techniques for modeling programme structures, algorithms, flow charts and pseudo codes.
- LO3: Understand how to use constants, variables, literals, primitive data types, and expressions.
- LO4: Learn how to use operators and operations, decision logic selection, simple selection and multiple selection.

Course Objectives

- LO5: Learn how to use functions, parameters and passing parameters.
- LO6: Learn how to write codes when solving simple problems.
- LO7: Understand and apply control structures and recursion
- LO8: Understand and apply simple data structures and file handling.
- LO9: Implement and debug applications to solve simple real life problems.

Topics

1. Overview and history of programming languages
2. Programming Fundamentals (Data types, Operators)
3. Control structures (Conditions and Loops)
4. Functions
5. Arrays and Strings
6. Pointers
7. Constructed Data Types
8. File Handling

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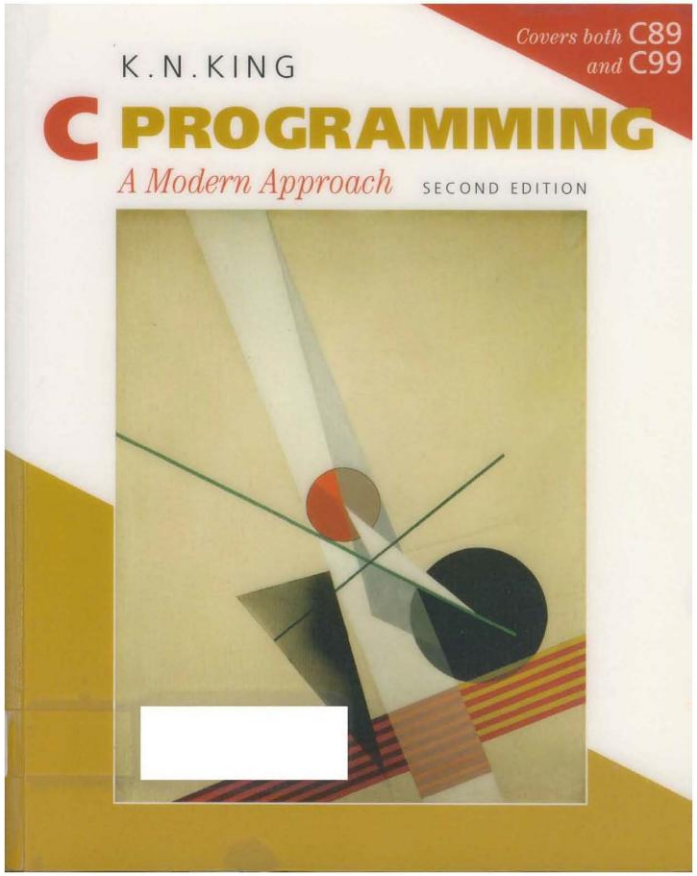
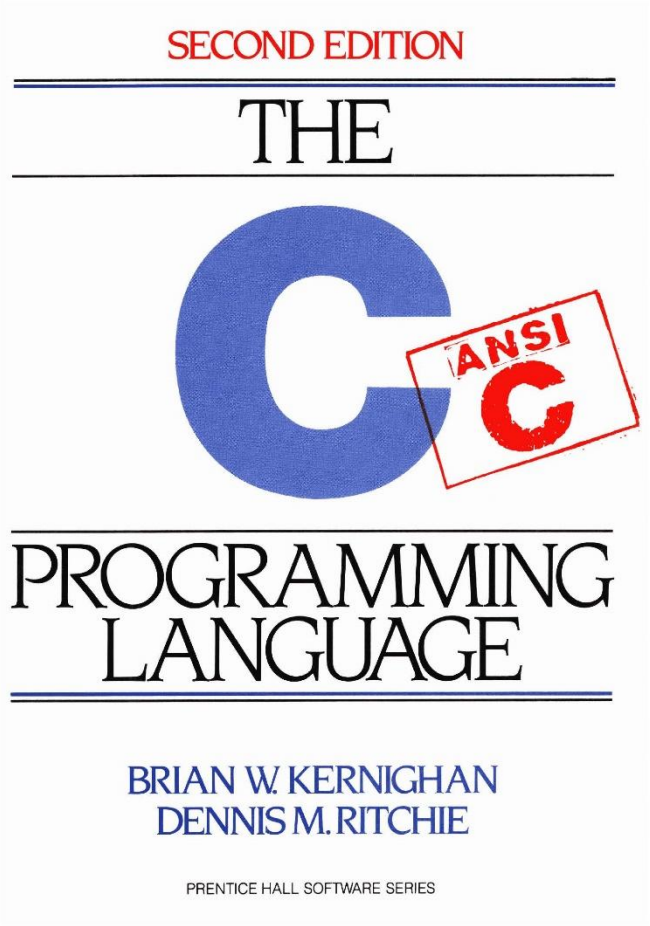
Roshan Sir

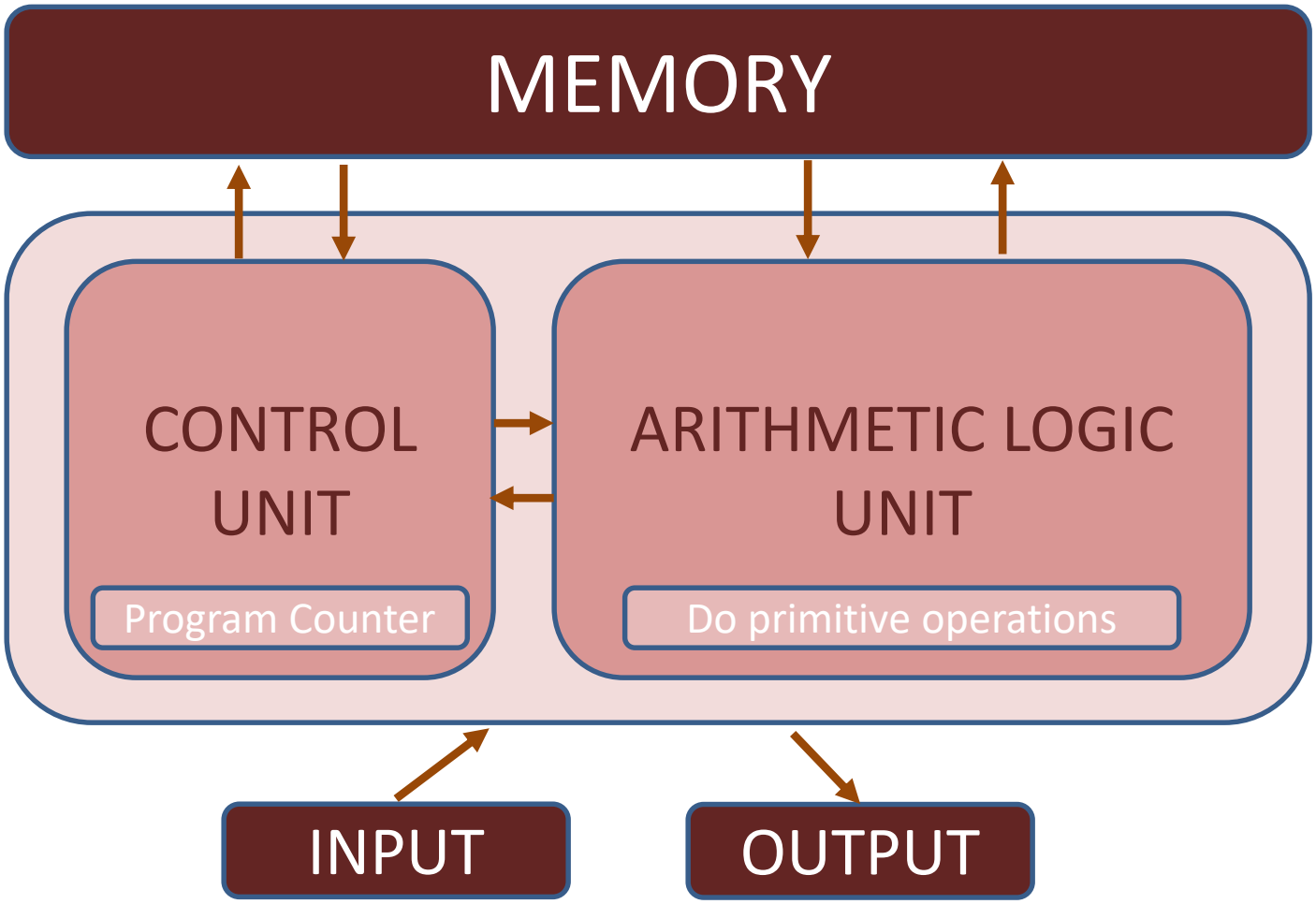
Reference Books



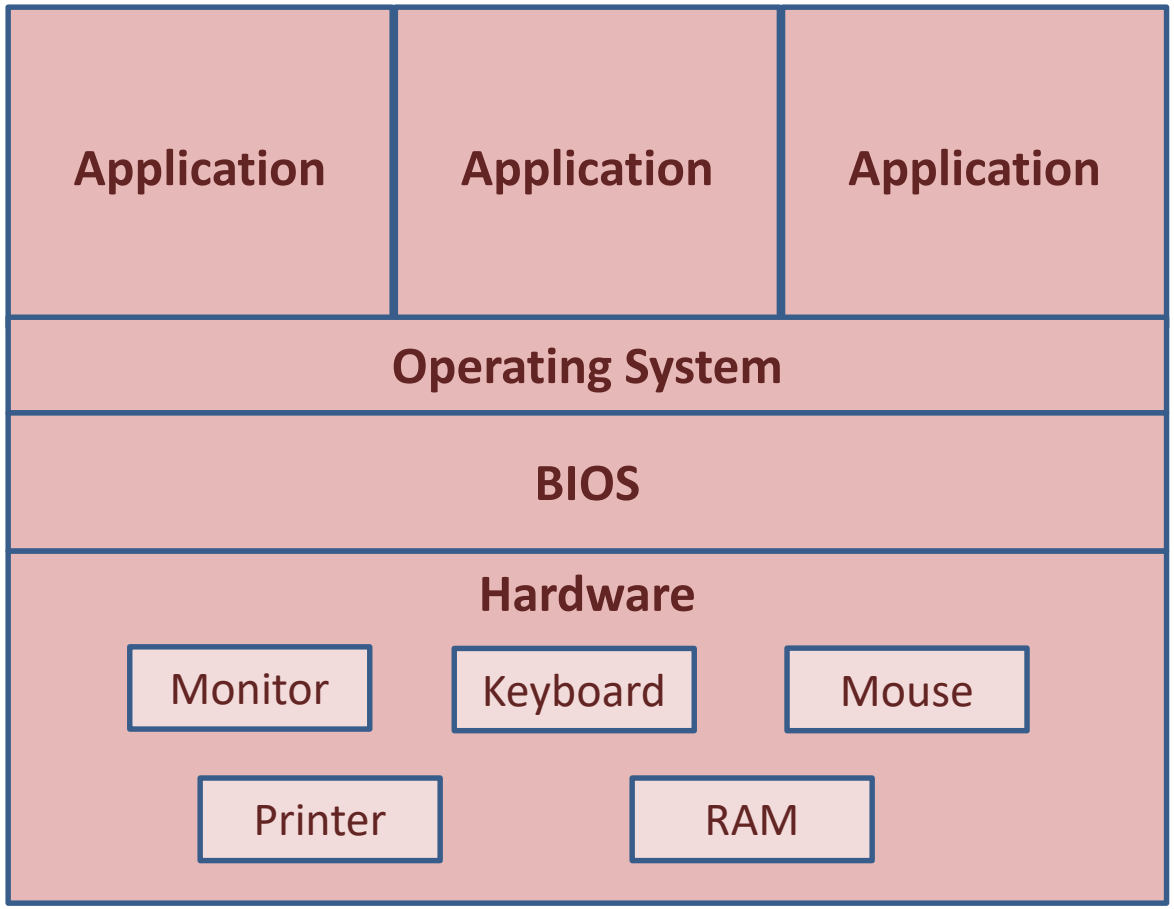


COMPUTERS & FLOW CHARTS

Computer is a Machine



Basic View of the Layered Architecture



Computers are Machines

- Sequence of instructions stored inside computer, built from predefined set of primitive instructions.
  1. Arithmetic & Logic
  2. Simple Tests
  3. Moving Data

- Special program executes each instruction in order.
- Use tests to change flow of control through sequence and stop when done.

What does a Computer do?

- Fundamentally:
  - Performs calculations
    - A billion of calculations per second!
- Remembers results
  - Thousands of gigabytes of storage!

- What kinds of calculations?
  - Built-in to the language
  - Ones that you define as the programmer.

Computers only know what you tell them

Computer Science?

- Computer science is no more about **computers** than astronomy is about telescopes.

- Instead, Computer Sconce is about **<u>Algorithms</u>**.

- An algorithm is a **formal specification** for stating a method to **solve a problem**.

Algorithm

- It is a formula, a recipe or a step by-step procedure to be followed in order to obtain the solution to a problem.
- To be useful as a basis for writing program.
- The algorithm must;
  - Arrive at a correct solution within a finite time.
  - Be clear, precise and unambiguous.
  - Be in a format which lends itself to an elegant implementation in a programming language.

Control Structures

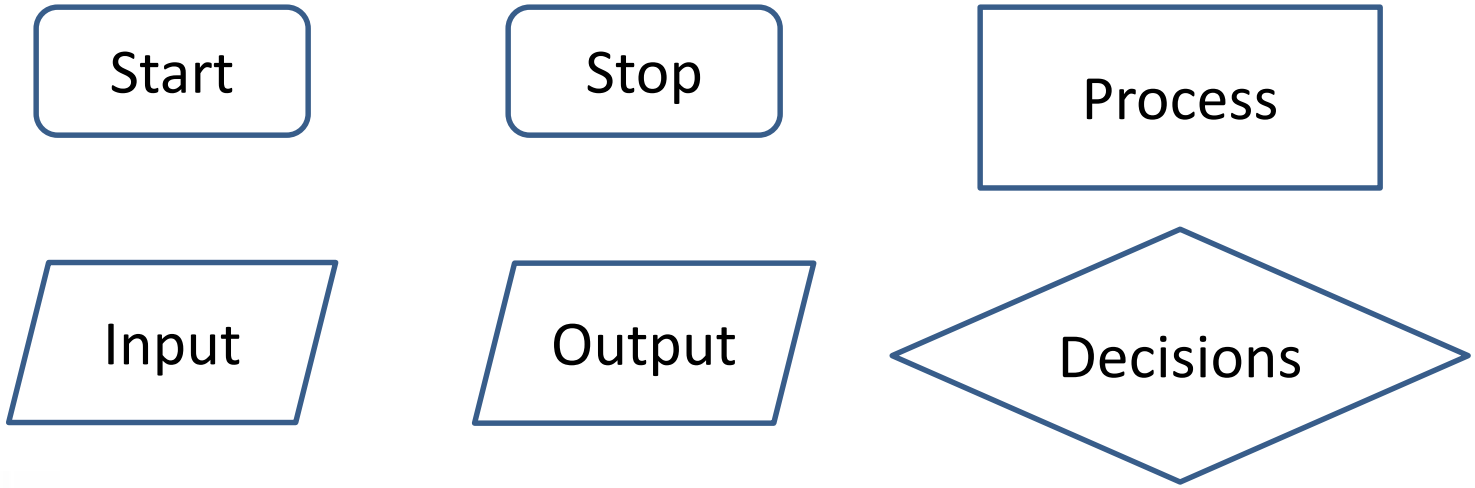
- The key to elegant algorithm design lies in limiting the control structures to only three constructs.

1. Sequence
2. Iteration
3. Selection

Flow Charts

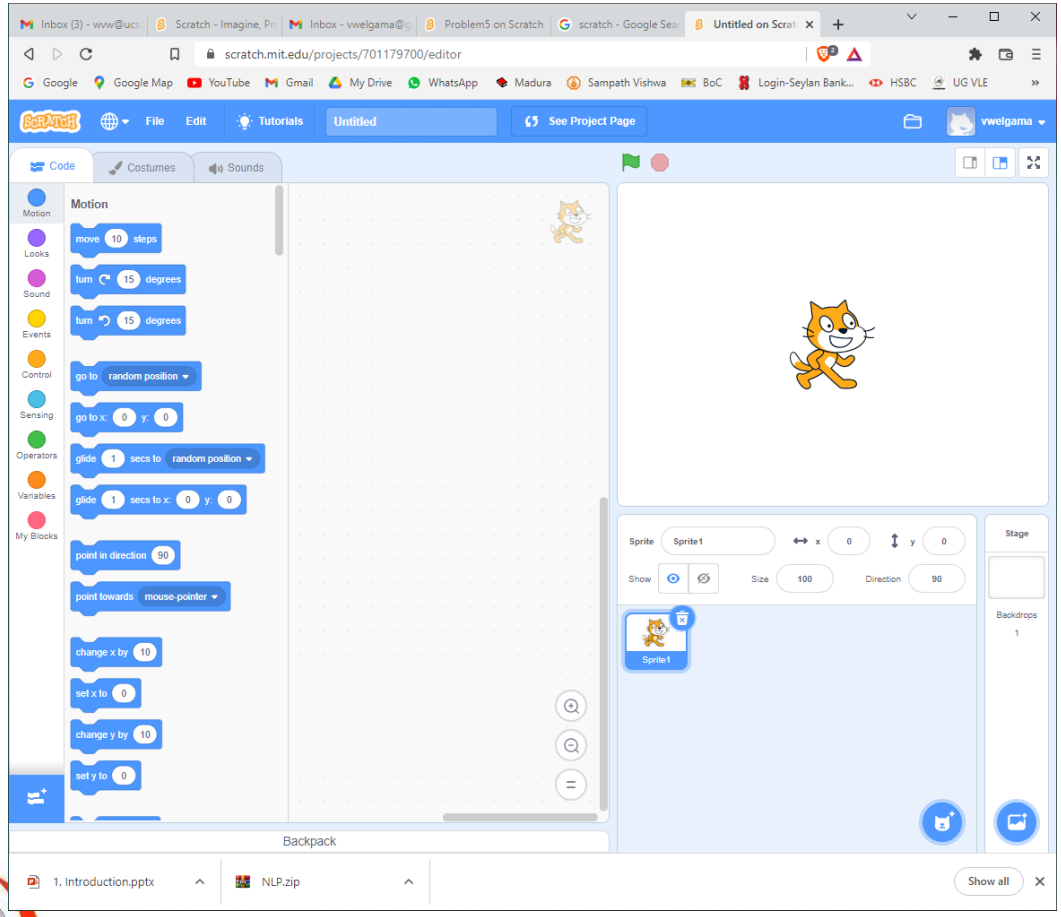
- Flow charts can be used as a way of expressing algorithms.

- Note that there are standard symbols to indicate:



SCRATCH

Scratch.mit.edu



Scratch

High-level programming language

Scratch is a high-level block-based visual programming language and website aimed primarily at children as an educational tool for programming, with a target audience of ages 8 to 16. Users on the site, called Scratchers, can create projects on the website using a block-like interface. Wikipedia

Influenced: ScratchJr

OS: Microsoft Windows, macOS, Linux (via renderer), HTML5, iOS, iPadOS, and Android

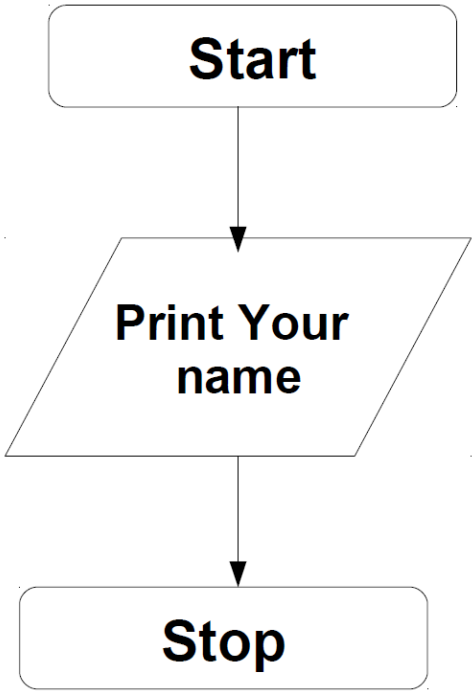
License: GPLv2 and Scratch Source Code License

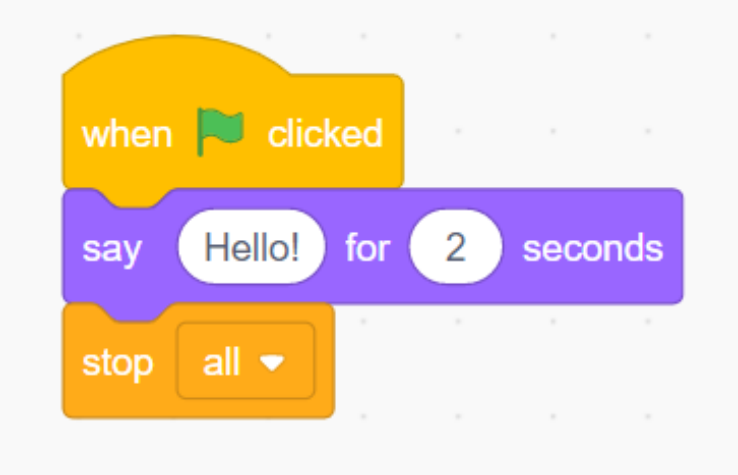
Filename extensions: .scratch (Scratch 0.x); *.sb, *.sprite (Scratch 1.x); *.sb2, *.sprite2 (Scratch 2.0); *.sb3, *.sprite3 (Scratch 3.0)

Paradigm: Event-driven, block-based programming language

Problem 1:

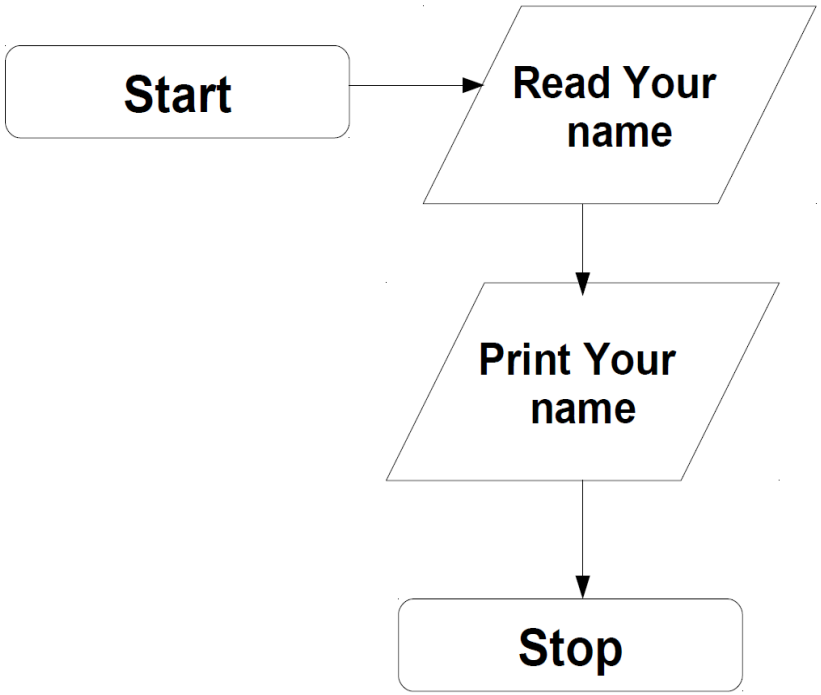
Write a program to print your name.

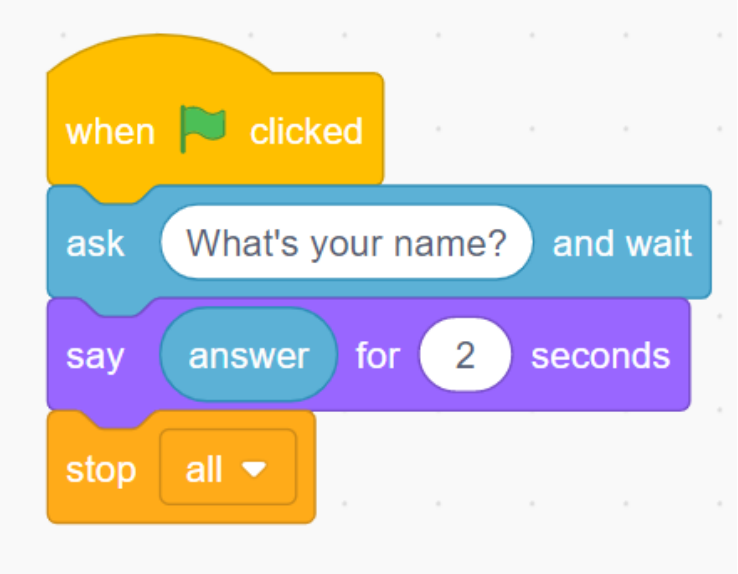




Problem 2:

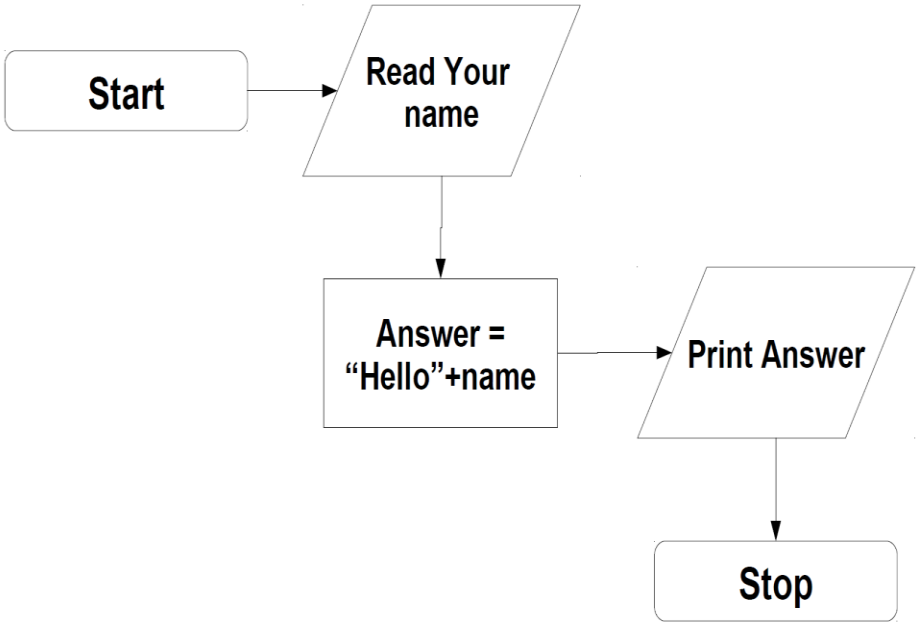
Write a program to read and print your name.

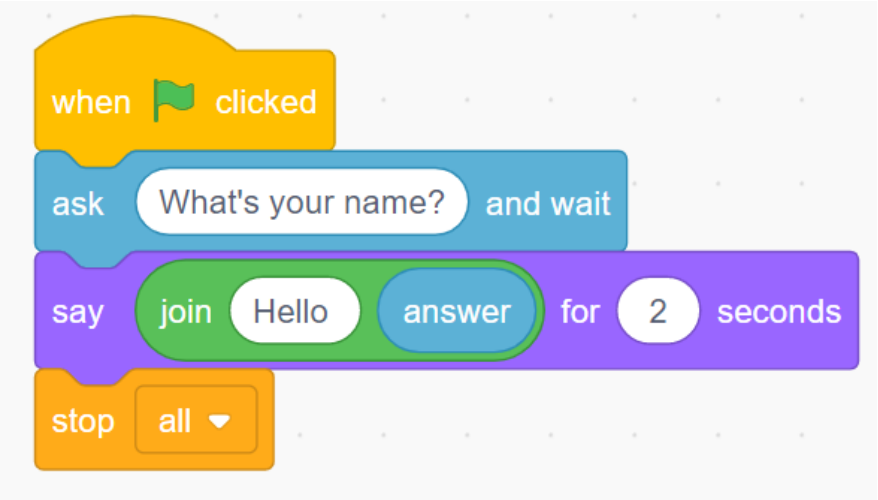




Problem 3:

Write a program to read and say “Hello” to your name

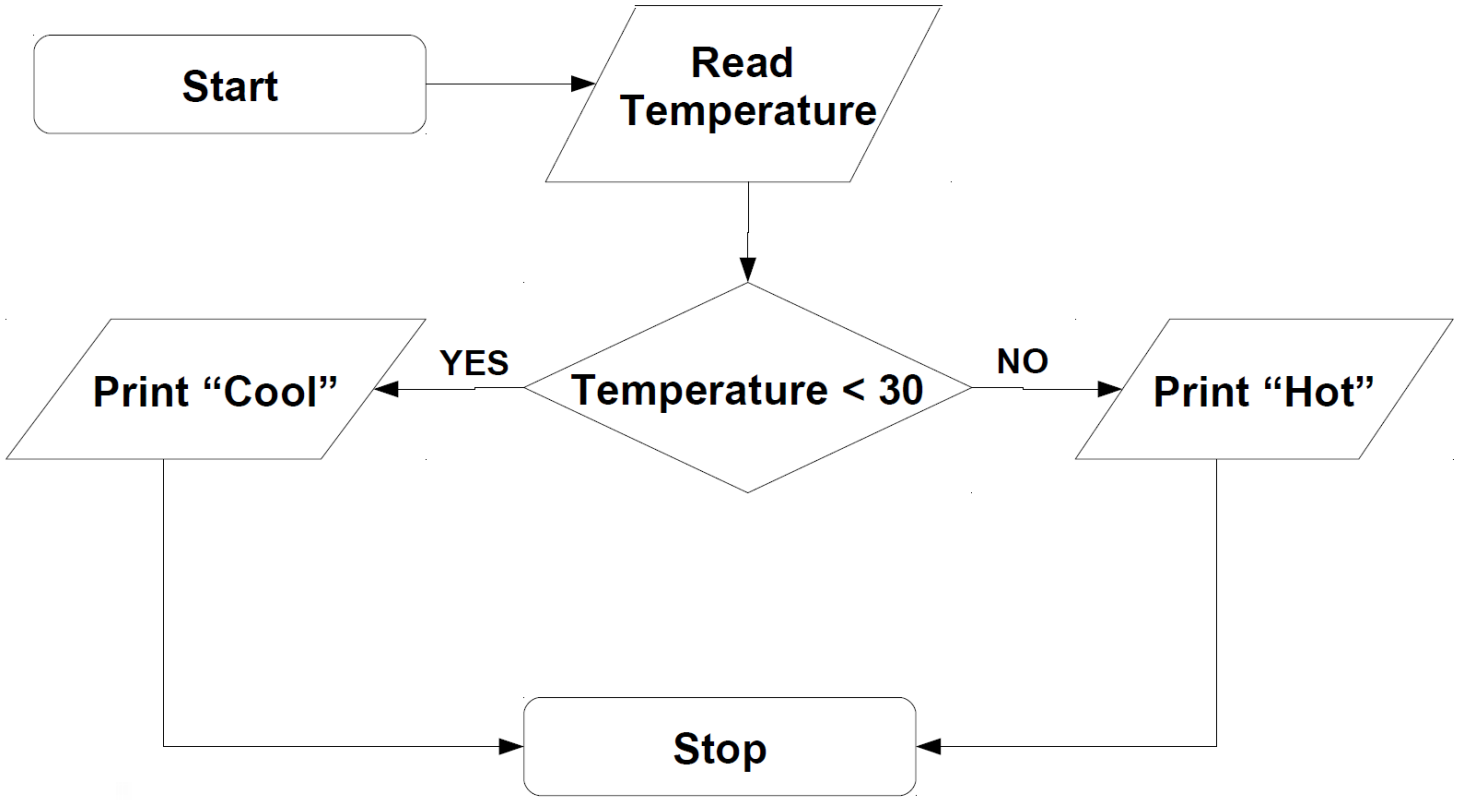




Problem 4:

Write a program to read temperature and print “Cool” or “Hot”

- If temperature is less than 30: Cool Otherwise: Hot



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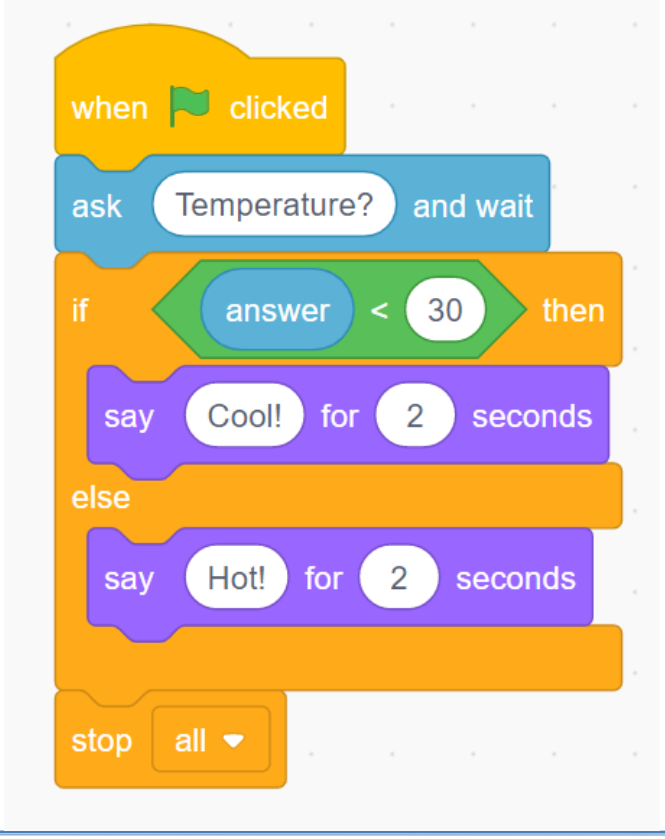
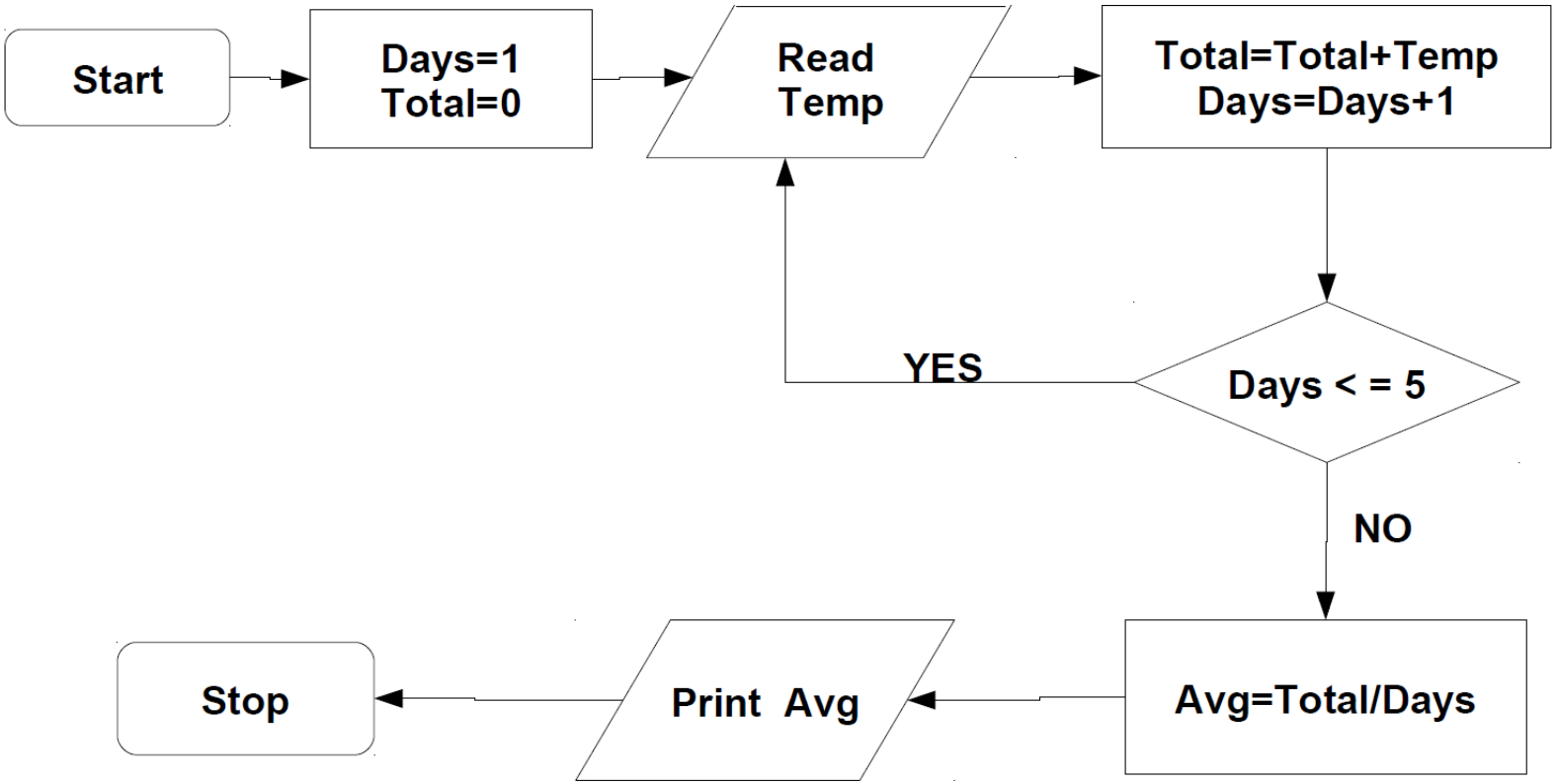


Problem 5:

Read temperature for 5 days and print average.



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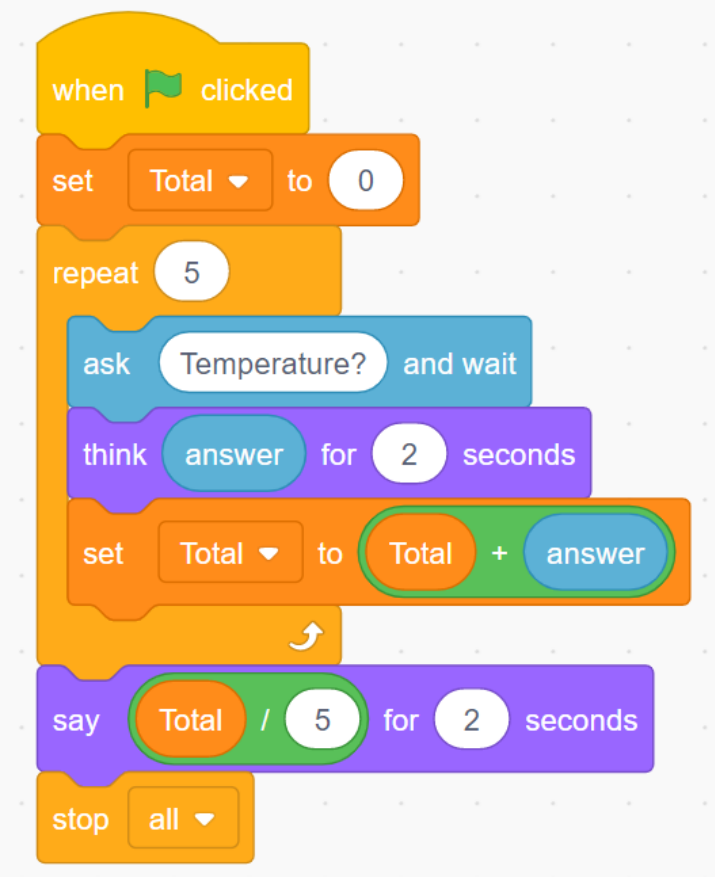
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